

Pharmacy Inventory Prediction System using Machine Learning

Abstract

Pharmacies must maintain the correct balance between medicine availability and stock cost. Overstocking causes medicine expiry and financial loss, while understocking leads to shortages and customer dissatisfaction. This project proposes a Machine Learning based Pharmacy Inventory Prediction System that predicts medicine demand using historical pharmacy sales data. The system uses a regression model to estimate daily medicine demand and calculate future inventory requirements. A Streamlit web dashboard provides an interactive interface for pharmacists to visualize and predict demand.

Keywords

Machine Learning, Inventory Management, Demand Forecasting, Pharmacy Analytics, Streamlit Dashboard

1. Introduction

Inventory management is one of the most important operations in pharmacies and medical stores. Medicines have limited shelf life and demand varies depending on season, disease outbreaks, and customer needs. Manual inventory management often leads to stock shortages or expired medicines.

Machine Learning techniques can analyze historical sales patterns and predict future demand. This project develops a Pharmacy Inventory Prediction System that uses historical sales data to forecast medicine demand. The system helps pharmacies maintain optimal stock levels and improve inventory planning.

2. Problem Statement

Pharmacies often face the following challenges:

- Difficulty predicting future medicine demand
- Overstocking leading to expired medicines
- Shortages of essential medicines

- Lack of data driven inventory management

The goal of this project is to develop a predictive system that forecasts medicine demand using machine learning techniques.

3. Objectives

The main objectives of this project are:

- Develop a machine learning model for predicting medicine demand
 - Reduce stock shortages in pharmacies
 - Avoid overstocking and medicine expiry
 - Provide a user friendly dashboard for pharmacists
 - Improve pharmacy inventory planning
-

4. Scope of the Project

The system predicts future demand for medicines using historical sales data.

Scope includes:

- Medicine demand prediction
- Inventory forecasting
- Visualization of demand trends
- Web dashboard for prediction

Future Scope:

- Integration with real pharmacy databases
 - Expiry date prediction
 - Multi branch pharmacy inventory system
 - AI based medicine recommendation
-

5. Literature Survey

Several studies show that machine learning algorithms can effectively predict demand patterns in retail and healthcare sectors. Regression algorithms are widely used for demand forecasting because they identify relationships between variables such as price, time, and product demand.

Modern analytics tools such as Python libraries and data science frameworks allow businesses to build predictive models that automate decision making. Streamlit has also made it easy to build interactive dashboards for machine learning applications.

6. System Architecture

The system architecture consists of the following components:

1 Data Collection

2 Data Preprocessing

3 Model Training

4 Model Storage

5 Prediction Module

6 Dashboard Interface

Workflow:

Dataset → Data Processing → Machine Learning Model → Model Storage → Prediction → Dashboard

7. System Requirements

Hardware Requirements

Processor : Intel i3 or higher

RAM : Minimum 4GB

Storage : 1GB free space

Software Requirements

Operating System : Windows / Linux / Mac

Programming Language : Python

Libraries : pandas, numpy, scikit-learn, matplotlib, joblib

Framework : Streamlit

IDE : Visual Studio Code

8. Dataset Description

The dataset used in this project contains synthetic pharmacy sales data.

Attributes include:

Medicine Name

Quantity Sold

Stock Available

Price

Date

This dataset helps train the machine learning model to understand sales patterns and forecast future demand.

9. Data Preprocessing

Data preprocessing prepares the dataset for machine learning.

Steps include:

- Loading the dataset
- Handling missing values
- Encoding categorical variables
- Feature selection
- Splitting data into training and testing sets

Label encoding converts medicine names into numeric values for the model.

10. Machine Learning Model

The system uses a regression algorithm to predict medicine demand.

Algorithm Used:

Linear Regression

Model Training Steps:

1 Load dataset

2 Prepare input features

3 Train regression model

4 Evaluate accuracy

5 Save trained model

11. Model Evaluation

Model performance is evaluated using regression metrics.

Mean Absolute Error (MAE)

R Squared Score

These metrics determine how accurately the model predicts demand.

12. Model Saving

After training, the model and preprocessing objects are stored as serialized files.

Files Generated:

model.pkl

label_encoder.pkl

medicine_price.pkl

These files allow the prediction system to load the trained model without retraining.

13. Prediction Module

The prediction module allows the user to enter a medicine name and number of future days.

The system predicts daily demand and calculates the required inventory for the selected period.

Steps:

- 1 Load trained model
 - 2 Encode medicine name
 - 3 Retrieve medicine price
 - 4 Predict daily demand
 - 5 Calculate total demand
-

14. Streamlit Dashboard

The project includes a Streamlit web dashboard that provides an interactive interface.

Dashboard Features:

- Select medicine
- Choose prediction days
- View demand prediction
- Visualize demand chart

The dashboard makes the system easy to use for pharmacists.

15. Advantages of the System

- Reduces medicine shortages
- Prevents overstocking
- Improves inventory planning
- Provides data driven decision making

- Easy web interface
-

16. Limitations

- Uses synthetic dataset
 - Limited prediction variables
 - No real time pharmacy integration
-

17. Future Enhancements

- Time series forecasting models
 - Integration with pharmacy billing systems
 - Expiry date prediction
 - Multi pharmacy inventory management
 - AI powered stock recommendations
-

18. Algorithms Used

Linear Regression Algorithm

Linear regression predicts the relationship between independent variables and a dependent variable.

Formula:

$$Y = a + bX$$

Where

Y = Predicted demand

X = Input features

b = slope

a = intercept

19. Project Folder Structure

pharmacy_inventory_prediction

- |— dataset.csv
- |— train_model.py
- |— predict_inventory.py
- |— visualization.py
- |— app.py
- |— model.pkl
- |— label_encoder.pkl
- |— medicine_price.pkl
- |— requirements.txt
- |— README.md

20. Source Code

train_model.py

```
import pandas as pd
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import LabelEncoder
import joblib

# Load dataset
data = pd.read_csv("dataset.csv")

# Encode medicine names
le = LabelEncoder()
data['medicine_encoded'] = le.fit_transform(data['medicine_name'])

# Create price dictionary
medicine_price = dict(zip(data['medicine_name'], data['price']))

X = data[['medicine_encoded', 'price']]
y = data['quantity_sold']
```

```
model = LinearRegression()
model.fit(X,y)

joblib.dump(model, 'model.pkl')
joblib.dump(le, 'label_encoder.pkl')
joblib.dump(medicine_price, 'medicine_price.pkl')

print("Model trained and saved")
```

predict_inventory.py

```
import joblib
import pandas as pd

model = joblib.load('model.pkl')
le = joblib.load('label_encoder.pkl')
medicine_price = joblib.load('medicine_price.pkl')

medicine = input("Enter medicine name: ")
days = int(input("Enter number of future days: "))

if medicine not in medicine_price:
    print("Medicine not found")
    exit()

price = medicine_price[medicine]
encoded = le.transform([medicine])[0]

input_data = pd.DataFrame({
    "medicine_encoded": [encoded],
    "price": [price]
})

daily_demand = model.predict(input_data)[0]

total_demand = daily_demand * days

print("Predicted demand:", int(total_demand), "units")
```

app.py (Streamlit Dashboard)

```
import streamlit as st
import joblib
import pandas as pd
```

```

model = joblib.load("model.pkl")
le = joblib.load("label_encoder.pkl")
medicine_price = joblib.load("medicine_price.pkl")

st.title("Pharmacy Inventory Prediction System")

medicine = st.selectbox("Select Medicine",list(medicine_price.keys()))

days = st.slider("Future Days",1,30,7)

encoded = le.transform([medicine])[0]

price = medicine_price[medicine]

input_data = pd.DataFrame({
    "medicine_encoded": [encoded],
    "price": [price]
})

prediction = model.predict(input_data)[0]

total = int(prediction * days)

st.write("Predicted Demand:",total,"units")

chart_data = pd.DataFrame({
    "Day": range(1,days+1),
    "Demand": [int(prediction)]*days
})

st.line_chart(chart_data.set_index("Day"))

```

21. Results

The machine learning model successfully predicts medicine demand based on price and medicine type. The Streamlit dashboard allows users to interactively view demand forecasts.

22. Conclusion

This project demonstrates how machine learning can improve pharmacy inventory management. By predicting future demand using historical data, pharmacies can maintain optimal stock levels and reduce financial loss caused by expired medicines.

23. References

Python Documentation

Scikit Learn Documentation

Streamlit Documentation

Machine Learning Research Papers